

Nice Prefixes

Tags:

Time Limit: 1 s

Memory Limit: 128 MB

Submission: 1

AC: 0

Score: 99.98

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Codes

AI Code Tips

Description

Consider strings formed from characters from an alphabet of size K . For example, if $K = 4$, our alphabet might be $\{a, b, c, d\}$, and an example string is $bbcac$.

For a string S , define $\text{count}(S, k)$ to be the number of occurrences of the symbol k in S . For example, $\text{count}(bbcac, b) = 2$ and $\text{count}(bbcac, a) = 1$.

A prefix of a string S is any string obtained from S by deleting some (possibly none) of the trailing characters of S . For example, the prefixes of acb are the empty string, a , ac , and acb .

A string S has "nice prefixes" if for every prefix P of S and for every two characters k_1 and k_2 in the alphabet, $|\text{count}(P, k_1) - \text{count}(P, k_2)| \leq 2$. For example, $bbcac$ has nice prefixes, but $abbbc$ does not because $\text{count}(abbb, b) = 3$ and $\text{count}(abbb, c) = 0$.

Count the number of strings of length L on an alphabet of size K that have nice prefixes. This number can be large, so print its remainder when divided by 1000000007.

Input

The input is a single line containing the two integers L and K , separated by spaces, with $1 \leq L \leq 10^{18}$ and $1 \leq K \leq 50$.

Output

Output a single line containing the number of strings of length L on an alphabet of size K that have nice prefixes, modulo 1000000007.

Samples

input	Copy
4 2	
output	Copy
12	

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:09:54